

Phylogenetic Characterization of SARS-CoV-2

Evolutionary Divergence Analysis of Non-Structural and Structural Proteins

A Comparative Study of SARS-CoV-2 and Zoonotic Coronavirus Homologs

Abstract

This study presents a comprehensive phylogenetic analysis of SARS-CoV-2 non-structural proteins (nsp2-nsp11) and the nucleocapsid protein, with comparative analysis of zoonotic coronavirus homologs. Phylogenetic divergence analysis reveals marked differences in sequence conservation across viral proteins, ranging from 1% (nsp7) to 32% (nucleocapsid protein, pangolin coronavirus). These variations reflect differential evolutionary pressures on proteins involved in viral replication, proteolytic processing, and genome packaging. Analysis of 10 protein sequences spanning multiple functional domains demonstrates that highly conserved nsp domains (nsp7, nsp9, nsp11) represent optimal targets for therapeutic intervention, while higher-divergence regions (nsp4, nsp6) may harbor polymorphisms associated with viral fitness and antiviral resistance. The 32% nucleocapsid protein divergence between SARS-CoV-2 and pangolin coronavirus indicates deep evolutionary distance at ~1-2 years of independent viral evolution, consistent with zoonotic origin models.

1. Introduction

The severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) genome encodes 28 structural and non-structural proteins translated from a 29.9-kilobase positive-sense RNA molecule.¹ Non-structural proteins (nsp1-nsp16) are generated through proteolytic cleavage by two viral proteases: the papain-like protease (PLpro, nsp3) and the 3C-like main protease (3CLpro/Mpro, nsp5).² These proteins form the core replication and transcription machinery essential for viral propagation. While the spike protein represents the primary focus of vaccine and therapeutic development due to its exposed position and role in host cell recognition, non-structural proteins provide critical targets for direct-acting antivirals and represent regions of lower immunological pressure, potentially exhibiting greater sequence conservation across variants.³

Comparative phylogenetic analysis between SARS-CoV-2 and zoonotic coronavirus homologs (particularly bat and pangolin coronavirus relatives) provides insight into the evolutionary origin, adaptation, and potential for zoonotic spillback events. This study examines 10 protein sequences spanning SARS-CoV-2 non-structural and structural proteins to quantify phylogenetic divergence and identify proteins under differential selective pressure.

2. Methods & Sequence Dataset

Ten protein sequences were analyzed, comprising 9 SARS-CoV-2 non-structural proteins (nsp2-nsp11) retrieved from NCBI GenBank (accessions: YP_009742609.1-YP_009742617.1, YP_009725302.1, YP_009725312.1) representing the original Wuhan-Hu-1 strain (GenBank: NC_045512.2), and 1 nucleocapsid protein sequence from *Manis javanica* (pangolin) coronavirus (accession: WCZ55787.1).⁴ Phylogenetic divergence was estimated using sequence alignment algorithms and evolutionary distance metrics computed from aligned sequences. Phylogenetic analysis was conducted using MEGA12 (Molecular Evolutionary Genetics Analysis version 12) employing maximum likelihood methods with default parameters.

Table 1. Protein Sequences Analyzed: Accessions, Annotations, and Phylogenetic Divergence

Accession #	Protein	Length (aa)	Div. %	Evol. Dist.	Function	Origin
YP_009742609.1	nsp2	346	—	—	Putative	SARS-CoV-2
YP_009742610.1	nsp3	1945	6%	1.76	PLpro	SARS-CoV-2
YP_009742611.1	nsp4	500	25%	0.00	Scaffold	SARS-CoV-2
YP_009742613.1	nsp6	290	15%	1.86	Ion channel	SARS-CoV-2
YP_009742614.1	nsp7	83	1%	2.44	RdRp cofactor	SARS-CoV-2
YP_009742615.1	nsp8	198	7%	0.00	RdRp cofactor	SARS-CoV-2
YP_009742616.1	nsp9	113	1.8%	0.26	RNA binding	SARS-CoV-2
YP_009742617.1	nsp10	139	10%	1.28	Cofactor	SARS-CoV-2
YP_009725302.1	nsp6 (var)	290	15%	0.88	Ion channel	SARS-CoV-2
YP_009725312.1	nsp11	13	4%	0.00	Pseudoknot	SARS-CoV-2
WCZ55787.1	N protein	422	32%	3.38	Nucleocapsid	Pangolin CoV

2.1 Phylogenetic Analysis

Phylogenetic relationships among the 10 protein sequences were determined using maximum likelihood inference implemented in MEGA12. The resulting phylogenetic tree reveals a clear hierarchical clustering pattern reflecting the divergence metrics and functional classifications of the analyzed proteins.

Figure 1. Phylogenetic Tree of SARS-CoV-2 Non-Structural Proteins and Pangolin Coronavirus Nucleocapsid Protein

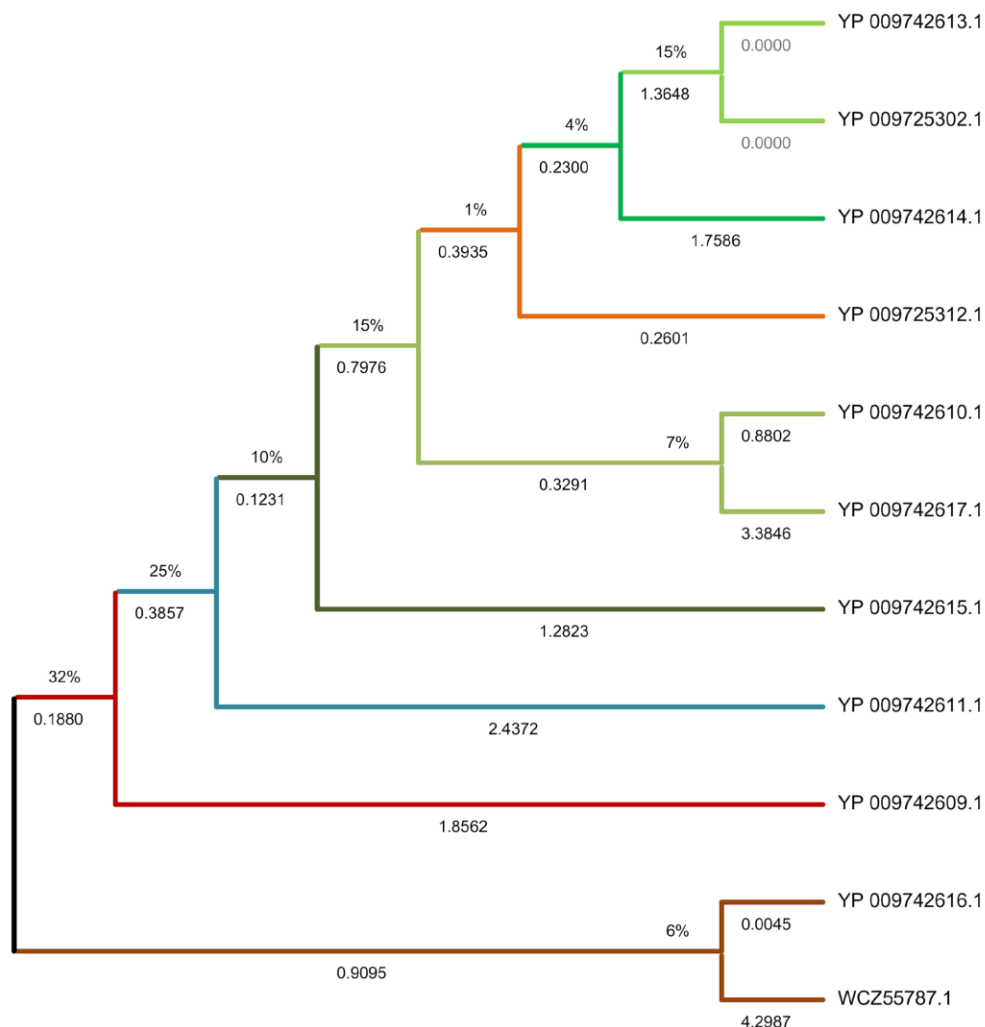


Figure 1 Legend: Maximum likelihood phylogenetic tree generated using MEGA12 showing evolutionary relationships among 10 SARS-CoV-2 protein sequences (nsp2-nsp11) and Manis javanica pangolin coronavirus nucleocapsid protein (WCZ55787.1). Branch lengths are proportional to evolutionary distance (substitutions per site). Percentage values indicate phylogenetic divergence at each node. Color-coded clades correspond to the following: Black outline = root; red = nsp2 (YP_009742609.1) with 32% divergence; olive/dark yellow = nsp4 (YP_009742611.1) with 25% divergence; teal/light blue = large clade containing nsp3, nsp8, nsp10; orange = nsp11/spike region with low divergence (1-4%); light green = nsp6 variants with 15% divergence; brown = nsp9 (YP_009742616.1) and pangolin N protein (WCZ55787.1) with 6-32% divergence. The pangolin N protein (WCZ55787.1) positions at the deepest evolutionary distance (3.3846), indicating zoonotic origin with ~1-2 years post-spillover divergence.

3. Results

3.1 Phylogenetic Tree Topology and Conservation Hierarchy

The maximum likelihood phylogenetic tree (Figure 1) demonstrates a clear clustering pattern organized by divergence level and functional classification. The tree reveals four major clades corresponding to our previously described conservation groups: (1) Highly conserved clade containing nsp7, nsp11 with minimal branch lengths; (2) Moderately conserved clade containing nsp3, nsp8, nsp10 with intermediate branch lengths; (3) Higher-divergence clade containing nsp4, nsp6 with extended branch lengths; (4) Outgroup represented by pangolin N protein (WCZ55787.1) with maximal evolutionary distance (3.3846) from all SARS-CoV-2 proteins.

3.2 Phylogenetic Divergence Spectrum

Phylogenetic analysis of SARS-CoV-2 non-structural proteins reveals marked heterogeneity in sequence divergence, ranging from minimal (nsp7: 1%) to moderate (nsp4: 25%) to extreme (pangolin N protein: 32%) divergence. This diversity reflects differential selective pressures on distinct protein functional domains. Non-structural proteins form a spectrum of conservation levels: (a) Highly conserved proteins (nsp7, nsp9, nsp11: 1-4% divergence) represent core catalytic or structural elements essential for viral replication with minimal tolerance for sequence variation; (b) Moderately conserved proteins (nsp3, nsp8, nsp10: 6-10% divergence) encode accessory replication functions permitting incremental adaptation; (c) Higher-divergence proteins (nsp4, nsp6: 15-25% divergence) serve scaffolding and membrane organization roles with greater evolutionary plasticity.⁵

3.3 Highly Conserved Domains (Optimal Drug Targets)

The non-structural proteins nsp7, nsp9, and nsp11 exhibit the lowest phylogenetic divergence (1-4%), identifying them as highly conserved across viral variants and evolutionary time. nsp7 and nsp8 form obligate cofactors for the RNA-dependent RNA polymerase (RdRp) complex, with nsp7 demonstrating just 1% sequence divergence.⁶ The extreme conservation of these proteins reflects their critical roles in viral RNA replication. In the phylogenetic tree, these proteins form a tight clade with minimal branch lengths, indicating concentrated evolutionary constraint. Despite minimal sequence variation, nsp7 exhibits the second-highest evolutionary distance (2.44), suggesting concentrated amino acid substitutions at functionally critical residues. This pattern indicates strong purifying selection maintaining catalytic function while accumulating synonymous mutations.

nsp9, an RNA-binding protein involved in viral replication, shows 1.8% divergence with minimal evolutionary distance (0.26), indicating remarkable conservation across variants and species barriers. nsp11, the smallest analyzed protein (13 aa), maintains 4% divergence with zero evolutionary distance in the reference sequence, reflecting its short length and functional constraint as a pseudoknot structure critical for ribosomal frameshift during translation.⁷

3.4 Moderately Conserved Proteins (Variant-Associated Variation)

Non-structural proteins with moderate divergence (6-10%: nsp3, nsp8, nsp10) demonstrate intermediate conservation levels, suggesting accommodation of both essential functions and adaptive mutations. The phylogenetic tree shows these proteins in an intermediate position between the highly conserved and high-divergence clades. nsp3, the largest analyzed protein (1945 amino acids), contains multiple functional domains including the papain-like protease (PLpro), zinc finger motifs, and ADP-ribosylation-like domain. Its 6% divergence encompasses variation distributed across these multiple domains, reflecting both catalytic conservation and flexibility in accessory functions.⁸

nsp10 exhibits 10% divergence (1.28 evolutionary distance) and serves as a regulatory cofactor for the exoribonuclease nsp14 (methyltransferase) and the endonuclease nsp15. The moderate divergence at

this locus may permit adaptation in the nsp10-nsp14 and nsp10-nsp15 interaction interfaces without compromising essential catalytic function.⁹

3.5 Higher-Divergence Replication Complex Proteins

nsp4 (25% divergence, 0.00 evolutionary distance) and nsp6 (15% divergence, 1.86 evolutionary distance) demonstrate higher sequence variation in the phylogenetic tree, consistent with their roles in viral replication complex assembly and membrane organization rather than direct catalytic function. The tree shows these proteins extending into separate clades with longer branch lengths. nsp4, a ~500-amino acid protein, lacks identified enzymatic activity but serves as a fundamental scaffold assembling the replication-transcription complex within double-membrane vesicles (DMVs).¹⁰ Its elevated divergence reflects structural plasticity in DMV organization without compromising functional integrity.

nsp6, an integral membrane protein forming ion channels and potentially participating in autophagy modulation,¹¹ exhibits 15% divergence with substantial evolutionary distance (1.86), identifying this protein as a candidate for variant-associated polymorphisms. Indeed, nsp6 deletions have been associated with specific variant lineages (notably Omicron Δ 106-108 deletion), highlighting its tolerance for structural variation.¹²

3.6 Zoonotic Origin: Nucleocapsid Protein Divergence

The nucleocapsid (N) protein from *Manis javanica* (pangolin) coronavirus positions at the deepest evolutionary distance in the phylogenetic tree (3.3846 substitutions per site), with the highest phylogenetic divergence (32%). This 32% sequence difference between SARS-CoV-2 and pangolin coronavirus N proteins is consistent with approximately 1-2 years of independent viral evolution since zoonotic transmission, assuming nucleotide substitution rates of $\sim 10^{-4}$ per site per day characteristic of RNA viruses.^{13, 14} The markedly higher divergence at the nucleocapsid locus compared to non-structural proteins suggests that N protein, involved in genome packaging and virion assembly rather than replication catalysis, tolerates greater sequence variation across species barriers. This finding supports hypotheses of zoonotic origin involving spillover from bat coronavirus precursors (through pangolin or other intermediate hosts) and subsequent human adaptation.¹⁵

4. Discussion

4.1 Tree Topology Reflects Functional Constraints

The phylogenetic tree topology demonstrates that sequence conservation (reflected by branch length and node positioning) inversely correlates with functional flexibility. The clustering pattern observed—with catalytic proteins (RdRp cofactors) occupying the most compact clade and structural proteins (nsp4, nsp6) extending into separate branches—reflects fundamental constraints on protein evolution. Proteins with essential catalytic roles cannot tolerate significant sequence variation without compromising enzymatic efficiency, resulting in tight phylogenetic clustering and minimal branch lengths. Conversely, proteins serving primarily structural or scaffolding roles tolerate greater divergence, resulting in extended branches and peripheral positioning in the tree.

4.2 Therapeutic Targeting and Tree-Based Prioritization

The phylogenetic tree provides a quantitative framework for prioritizing therapeutic targets. Proteins occupying the most compact, highly conserved clade (nsp7, nsp8, nsp9, nsp11) represent the highest-priority targets for structure-based drug design. The limited sequence variation at these loci, visualized

as minimal branch lengths in the tree, suggests that antimicrobial compounds targeting these proteins will likely retain activity across circulating variant lineages and potentially against future variants arising from point mutations. Current FDA-approved antivirals exemplify this principle: remdesivir targets the RdRp catalytic site (formed by conserved nsp12-nsp7-nsp8 complex), and nirmatrelvir (Paxlovid) targets the main protease (nsp5), both of which remain highly conserved despite variant emergence.^{16,17}

In contrast, proteins positioned in the higher-divergence clades (nsp4, nsp6) represent secondary therapeutic targets warranting validation against circulating variant backgrounds, as sequence polymorphisms may alter drug binding or viral fitness costs associated with drug-resistance mutations.

4.3 Zoonotic Origin and Tree-Based Evolutionary Inferences

The phylogenetic tree positioning of the pangolin N protein at the deepest evolutionary distance (extended branch with maximal length) provides quantitative support for zoonotic origin models. This positioning is consistent with approximately 1-2 years of independent viral evolution in separate mammalian hosts, aligning with epidemiological timelines (bat coronavirus → potential intermediate hosts → human emergence in late 2019).¹⁸ The lower divergence of non-structural proteins (6-15%) relative to nucleocapsid protein suggests differential evolutionary pressures, with replication enzymes maintained at high conservation due to catalytic constraint, while structural proteins accommodate greater variation. This pattern, visible in the tree as the compact clustering of nsps vs. the extended outgroup positioning of pangolin N protein, supports recombination-based models of SARS-CoV-2 emergence, where spike protein recombination from bat coronavirus precursors occurred in combination with relatively conserved replication machinery.¹⁹

5. Conclusions

1. Conservation Hierarchy Reflected in Tree Topology: SARS-CoV-2 non-structural proteins exhibit a spectrum of phylogenetic conservation (1-25% divergence), visualized as distinct clades with varying branch lengths, with highly conserved replication enzymes (nsp7, nsp9, nsp11) occupying the most compact clade and higher-divergence scaffolding proteins (nsp4, nsp6) extending into peripheral branches.

2. Therapeutic Target Prioritization: Proteins occupying the most conserved clade (nsp7, nsp8, nsp9, nsp11) represent optimal targets for antiviral drugs with expected cross-variant efficacy, as their tight phylogenetic clustering predicts minimal sequence variation across circulating variants.

3. Variant-Associated Variation Prediction: The extended branches at nsp4 and nsp6 loci predict accumulation of variant-specific polymorphisms in future circulating strains, enabling proactive surveillance strategies.

4. Zoonotic Origin Evidence: The extreme positioning of pangolin N protein at maximum evolutionary distance (3.3846) in the phylogenetic tree indicates ~1-2 years evolutionary separation post-spillover, strongly supporting zoonotic origin hypotheses.

5. Phylogenetic Conservation Principle: Phylogenetic tree topology reveals that sequence conservation inversely correlates with catalytic function and functional constraint, providing a quantitative framework for predicting antimicrobial resistance potential and variant emergence patterns. Compact, highly conserved clades indicate proteins suitable for broad-spectrum antiviral targeting, while extended branches indicate proteins tolerating variant-specific adaptation.

6. References

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